

Life Expectancy Changes Due to Kidney Donation Across Race, Sex, and Age

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Abstract

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Keywords

keyword1, keyword2, keyword3

Methods

Model overview

We constructed a Markov state-transition model to compare lifetime outcomes for two hypothetical cohorts of otherwise identical individuals: one cohort that donates a kidney and one that does not. Both cohorts enter the model at the age of donation in full health, and the simulation advances in annual cycles until age 100. The analysis was conducted from the perspective of the individual donor rather than the health-care system; outcomes are reported as undiscounted remaining life-years, consistent with the standard approach for life-expectancy analyses¹. The base-case age at donation is 40 years; subgroup analyses span ages 25–55. We used a Monte Carlo simulation with 5×10^6 individuals per arm and cross-validated against a deterministic analytic cohort implementation.

Health states

The model contains seven mutually exclusive health states (Figure 1): *Healthy*, *ESRD onset*, *Dialysis*, *Standard waitlist*, *Priority waitlist*, *Post-transplant*, and *Dead* (absorbing). At ESRD onset, the cohort branches: a fraction is listed preemptively (bypassing dialysis entirely) and the remainder initiates haemodialysis. The donor arm routes ESRD survivors to the *Priority waitlist*, reflecting the allocation priority afforded to prior living donors (PLD) under KAS250; the non-donor arm routes to the *Standard waitlist*. Graft failure returns patients to the *Dialysis* state, bypassing the ESRD-onset branch.

ESRD risk after donation

Fifteen-year cumulative ESRD incidence was taken from Muzaale et al.²: 30.8 per 10,000 (95% CI 24.3–38.5) for donors and 3.9 per 10,000 (95% CI 0.8–8.9) for age-, sex-, and health-matched controls (8-fold relative risk). Race-specific donor rates were 74.7/10,000 (Black) and 22.7/10,000 (White)². The White non-donor baseline was taken from Grams et al.³ (sex-averaged 0.05%/15 yr), because zero ESRD events were recorded in White matched controls in Muzaale et al.

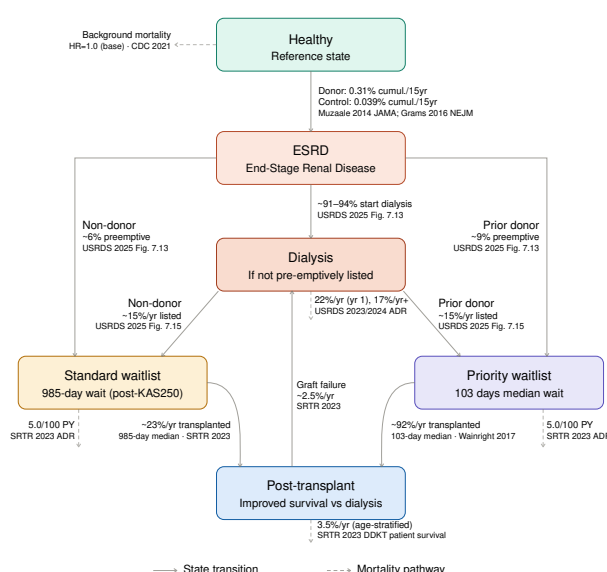


Figure 1. The Markov model for kidney donation.

ESRD hazard was modelled with a Weibull distribution (shape $k = 1.5$), calibrated via a competing-risk bisection algorithm to reproduce the 15-year cumulative incidence while accounting for background-mortality attrition in the at-risk pool. The accelerating hazard ($k > 1$) is supported by the Massie et al. 20-year cumulative incidence of 34 per 10,000 (vs. $\approx 21/10,000$ predicted under a constant-hazard model)⁴. Within-donor subgroup hazard ratios were drawn from Massie et al.: 2.96 (95% CI 2.25–3.89) for Black race, 1.88 (95% CI 1.50–2.35) for male sex, and 1.40 per decade (non-Black donors)⁴.

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Dialysis survival and waitlist listing

First-year haemodialysis mortality was 22% and subsequent-year mortality 17% ($\approx 42\%$ five-year survival), consistent with USRDS estimates⁵.

At ESRD onset, a fraction of patients is listed preemptively before dialysis initiation, avoiding dialysis mortality entirely. Preemptive listing probabilities were drawn from USRDS 2025 ADR Figure 7.13⁶: 5.8% overall for the non-donor arm and 9.4% for the donor arm (age 18–44 cohort, reflecting the younger and healthier donor-like population).

Among patients on dialysis, the conditional annual probability of joining the waitlist was set to 15%/year (base case), derived from USRDS 2025 ADR Figure 7.17⁶. Approximately 25% of 18–44 year-old ESRD patients are listed or transplanted within one year, of whom $\approx 9.4\%$ were preemptively listed, yielding an estimated post-dialysis listing rate of $\approx 15.6\%$ /year for the donor-like cohort (conservative base case: 15%/year; one-way sensitivity range 5–30%/year).

Transplant waiting time and waitlist outcomes

Transplant probability while listed was modelled as a time-homogeneous exponential process parameterised by the median waiting time, a standard approximation for waiting-time analyses⁷. For the standard waitlist, the post-KAS250 national median of 985 days was used⁸; for the PLD priority waitlist, the post-KAS median of 102.6 days was used⁹. Waitlist mortality was 5.0 deaths per 100 patient-years (SRTR 2023 ADR Figure KI 24) and competing removal 6.81%/year (SRTR 2023 ADR Figure KI 22)¹⁰.

Post-transplant survival

Post-transplant annual mortality was derived from age-stratified 5-year Kaplan–Meier estimates for DDKT recipients in the SRTR 2023 ADR (Figure KI 70, 2016–2018 cohort)¹⁰: 0.9%/year (ages 18–34), 1.8%/year (35–49), 3.9%/year (50–64), and 6.9%/year (≥ 65). The base case uses DDKT survival because prior living donors who develop ESRD predominantly receive transplants via the deceased-donor priority pathway rather than living-donor transplantation, consistent with USRDS data showing that living-donor transplants account for a small minority of transplants in this population^{11,12}. A one-way sensitivity analysis substitutes LDKT patient survival (SRTR 2023 ADR Figure KI 76: 0.4%, 0.8%, 1.7%, and 3.9%/year by age band) to bound the effect of this assumption. Annual graft failure was 2.5%/year post-year 1¹⁰; graft failure returned patients to the Dialysis state.

Background mortality and donor all-cause survival

Background mortality in the Healthy state was taken from the 2021 US Life Tables (CDC, National Vital Statistics Reports Vol. 72 No. 12)¹³, with sex-specific tables used in sex-stratified analyses. The base-case donor all-cause mortality hazard ratio versus matched non-donors is 1.0, consistent with US-based studies^{2,14} and the Grams et al. 2018 meta-analysis (pooled HR 0.984, 95% CI 0.743–1.302; 9 studies,

$n = 118,426$ donors)¹⁵. An HR of 1.30 (Mjøen et al.¹⁶, a Norwegian cohort in which 80% of donors were first-degree relatives of the recipient) was examined as the pessimistic sensitivity scenario.

Subgroup analyses

Age at donation Analyses were repeated at ages 25, 35, 40, 45, and 55 years at donation. Donor ESRD rates were scaled by the Massie 2017 age HR (1.40/decade, non-Black donors)⁴; non-donor baselines were held at the Muzaale overall matched-control rate.

Race Race-specific donor ESRD rates (Black 74.7/10,000; White 22.7/10,000) and race-specific non-donor baselines (Black sex-averaged 0.195%/15 yr; White sex-averaged 0.05%/15 yr) were taken from Muzaale et al.² and Grams et al., respectively³. Race-specific waitlist mortality and post-transplant survival were from SRTR 2023 Figures KI 25 and KI 71¹⁰.

Sex Sex-specific donor ESRD rates were derived by applying the Massie 2017 male-sex HR (1.88)⁴ to the overall donor rate, weighted by the observed donor sex mix (60% female; SRTR 2023 ADR Figure KI 7)¹⁰. For non-donor baselines, we used the Grams 2016 White sex-specific rates (female 0.04%/15 yr; male 0.06%/15 yr) rather than applying the within-donor sex HR to the control arm, because Massie 2017 hazard ratios describe variation within the donor cohort and are not appropriate for scaling a different reference population³. This sourcing rule, Muzaale matched-control rates for overall analyses and Grams direct rates wherever sex or race is stratified, was applied consistently across all matrix subgroup figures.

Probabilistic sensitivity analysis

We conducted a PSA with 200 parameter draws. Sampled quantities and their distributions were: ESRD 15-year cumulative risks (beta distributions parameterised from the Muzaale 2014 published 95% CIs); donor all-cause mortality HR (log-normal, $\mu = -0.016$, $\sigma = 0.143$, derived from the Grams 2018 meta-analysis CI); Weibull shape k (log-normal, $\sigma = 0.13$, representing structural uncertainty absent published CIs for this parameter); and waitlist listing probability (beta, fractional SE = 40%, consistent with the 0.05–0.30 plausible range). Registry-derived parameters (SRTR, USRDS) have negligible sampling error relative to structural uncertainty and were fixed at base-case values; scenario variation in these parameters is captured by the one-way sensitivity analysis. We report the mean Δ LE, 95% credible interval, and the probability that donation is life-expectancy-neutral or beneficial.

One-way sensitivity analysis

Each parameter was varied individually while others were held at base-case values. Key scenarios included: no priority access (PLD wait set to the standard 985-day median), donor all-cause mortality HR = 1.30 (Mjøen pessimistic estimate), ESRD relative risk $\times 11$ (Mjøen upper-bound ESRD HR), PLD wait 50 days (optimistic), post-transplant survival at LDKT quality (SRTR Figure KI 76), and dialysis mortality $\pm 50\%$.

Calibration

The model's transplant probability was validated against the SRTR 2023 3-year waitlist outcome distribution (Figure KI 22, 2019–2021 listing cohort)¹⁰. This comparison is non-circular because the transplant probability was derived from the median waiting time, not from the 3-year outcome distribution. The model predicted 44.3% transplanted at three years versus 44.3% observed (DDKT + LDKT combined; $\Delta = 0\%$). The waitlist removal rate was back-calculated via a competing-risk bisection algorithm to reproduce the SRTR-observed 19.1% removed at three years (12.6%/year); the naïve formula $1 - (1 - \text{CIF})^{1/3}$, which ignores competing depletion by transplantation and death, was not used. The model over-predicted three-year waitlist mortality (10.0% vs. 6.8% observed, $\Delta = +47\%$), likely reflecting the application of a 2023 cross-sectional mortality rate (confirmed 5.0/100 patient-years from the raw SRTR data file) to a 2019–2021 listing cohort; this bias is symmetric across both arms and conservative (slightly overstates the cost of the waitlist period for both donors and non-donors equally). No independent cohort exists tracing the complete pathway from living donation through ESRD onset to transplantation; these checks represent the available face-validity evidence for the model.

Analytic cycle length and discounting

Annual (one-year) cycles were used throughout. Life-years accumulated with an end-of-cycle convention; the resulting systematic underestimate of approximately 0.5 years applies equally to both arms and cancels in ΔLE . Future life-years were not discounted, consistent with the convention for pure life-expectancy analyses¹.

Software and reproducibility

All analyses were implemented in Python 3.10 (NumPy, pandas, matplotlib). The simulation, parameter assembly pipeline, sensitivity analyses, and calibration checks are fully reproducible from the public code repository at <https://github.com/rupumped/kidney-donor-le>.

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